Biology

Chapter 1.1: Carbohydrates

Welcome to AnithUncommon's **Sample Resources!**

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In Biology Chapter 1.1, you will explore one of the most important molecules in almost all living organisms.

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Kenneth V. to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology, join our nonprofit!

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This sample is made thanks to, Kenneth V.

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

- Describe the structure of Glucose, Amylose, Amylopectin, Cellulose, and Glycogen
- Explain how the bonds between Carbohydrate molecules form

Reflective Questions to Consider:

Carbohydrates are essential molecules in all living organisms. Thus, it is important to consider the following questions:

- Since it is essential to life, what exactly are the common carbohydrate molecules?
- How do the molecules maintain their structure in accordance with their functions?

Chapter 1: Biological Molecules	
Tip! • **It is imperative to understand the basics of organic chemistry before diving into this subject. Organic molecules are always composed of Carbon and Hydrogen atoms bonded covalently in some way.** **Activities in Class:** • **Lesson Breakdown** • **Drawing Quiz** • **Kahoot-type quiz**	
1.1 . Carbohydrates	**Mentor's Notes**

Glucose

A. **Description**

→ A glucose molecule is considered a **Monosaccharide** (Mono- meaning one, saccharide meaning sugar)

→ It is considered a Monosaccharide due to it being the basic unit of more complex saccharides containing Glucose

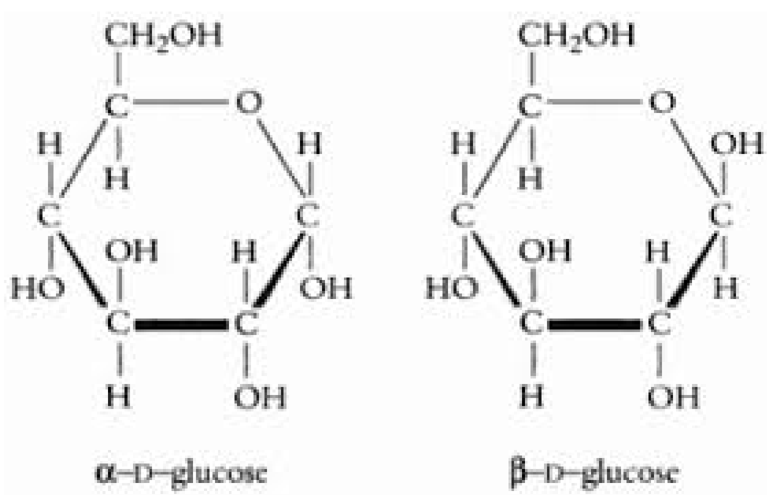
→ Glucose is a **hexose** molecule (Hex- meaning six, referring to its six carbon atoms)

→ Glucose is an **essential** molecule for **energy** in cells

B. **Structure**

→ Glucose technically has many structures due to its **isomerism** (molecules that share the same molecular formula but have different structures/arrangements)

→ The following two are the molecules used in this chapter: (Alpha [α], Beta [β])



(*Note the difference between the right-most* ***hydroxide (OH)****'s position*)

C. **Maltose**

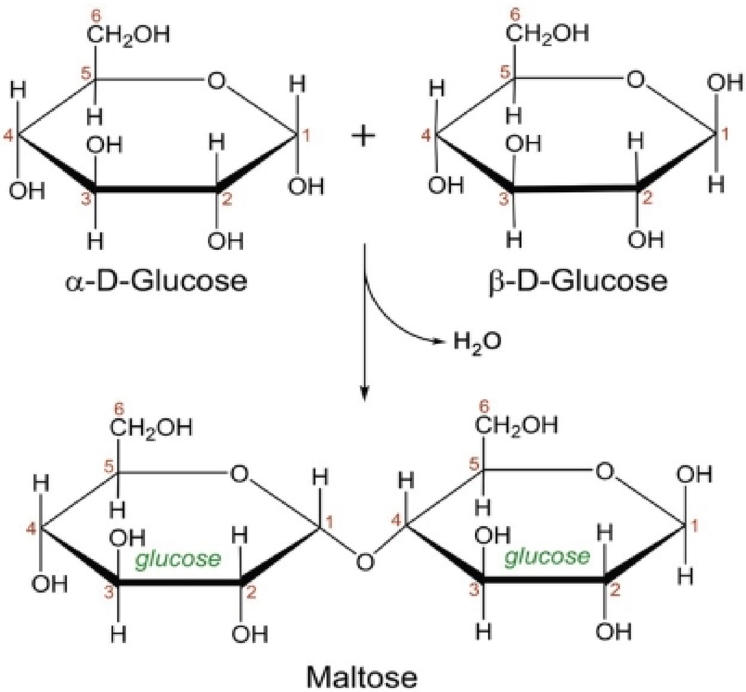
Biological molecules can be simply split into: (Ascending complexity)

- **Monomers**
- **Dinomers**
- **Polymers**
- **Macromolecules**

It's good to understand that the structures of the molecules are complementary to their use in a living organism. For example, Cellulose is straight chained to be able to form more strong hydrogen bonds between chains. Since cellulose is used to form the fibers of a plant cell's walls, the hydrogen bonds are crucial to maintain rigidity. The hexagon shape in glucose molecules is its most stable form.

→ When two glucose molecules "link", they undergo a reaction known as **condensation** that produces **Maltose** and **water** (the reverse of the reaction is called Hydrolysis which uses up water)

→ Maltose is considered a **disaccharide** (di- meaning two) due to having two separate glucose molecules bonded to each other



→ The bond that forms due to condensation is called a **glycosidic** bond (In the case of Maltose, it's referred to as a **1,4-Glycosidic bond**)

→ Maltose is more commonly found when breaking down Starch molecules during digestion

Starch

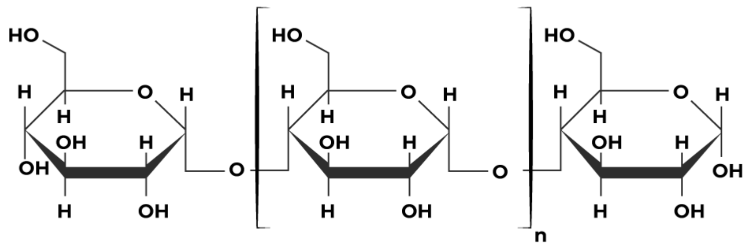
A. **Description**

→ Starch is considered to be a mixture of two **Polysaccharides** (Poly meaning many): Amylopectin and Amylose

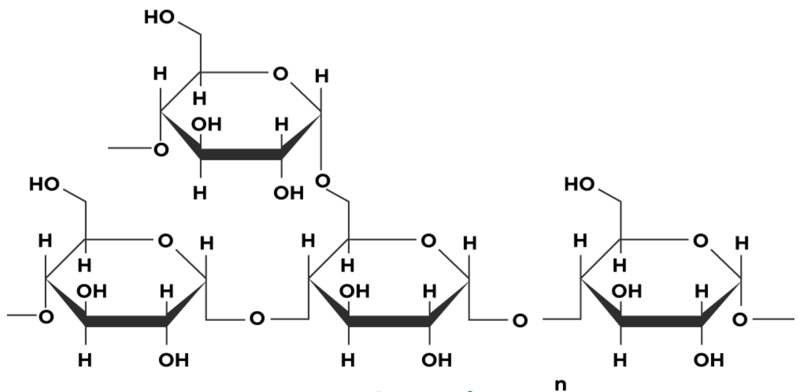
→ Both polysaccharides serve the purpose of **storing**

excess energy (glucose) to use later on

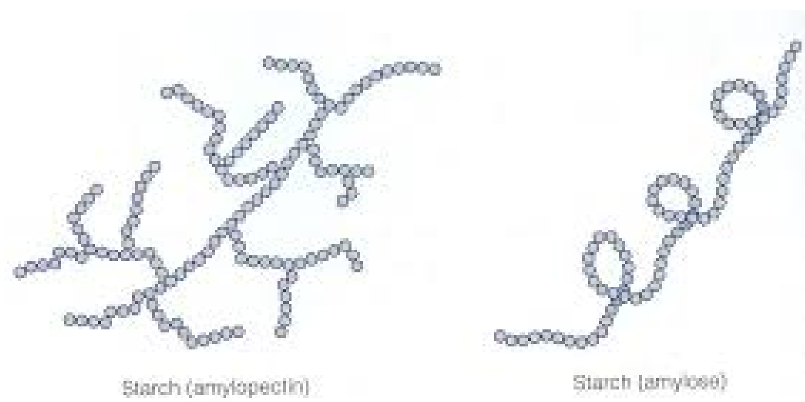
B. Structure



Amylose



Amylopectin



(Note that due to amylose's straight chain structure, it coils due to present Hydrogen Bonds)

(The branches of amylopectin has a 1,6-Glycosidic Bond)

C. Function

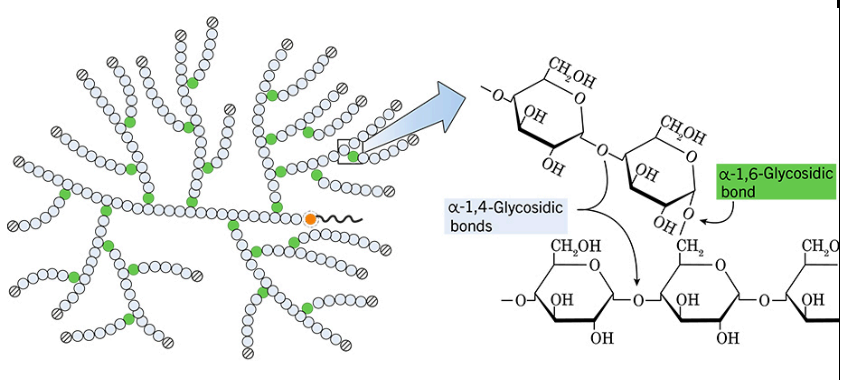
→ Starch is considered **insoluble** in liquid due to both of its polysaccharides having **strong hydrogen bonds**

→ Amylose is more **compact** due to its **helical shape**, perfect for **long-term energy storage**

→ Amylopectin is **branched**, increasing its **surface area** for **enzymes** to break down the glycosidic bonds holding the glucose molecules more easily

D. **Glycogen**

→ Similar to starch, Glycogen is a molecule meant for **storage** but in animals



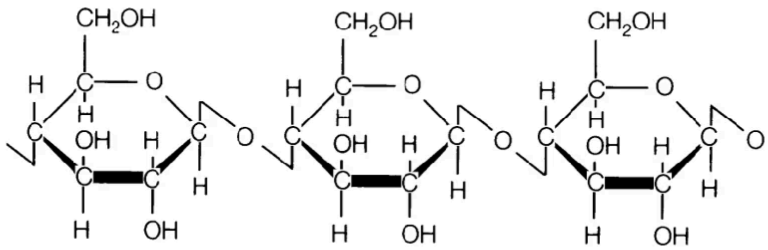
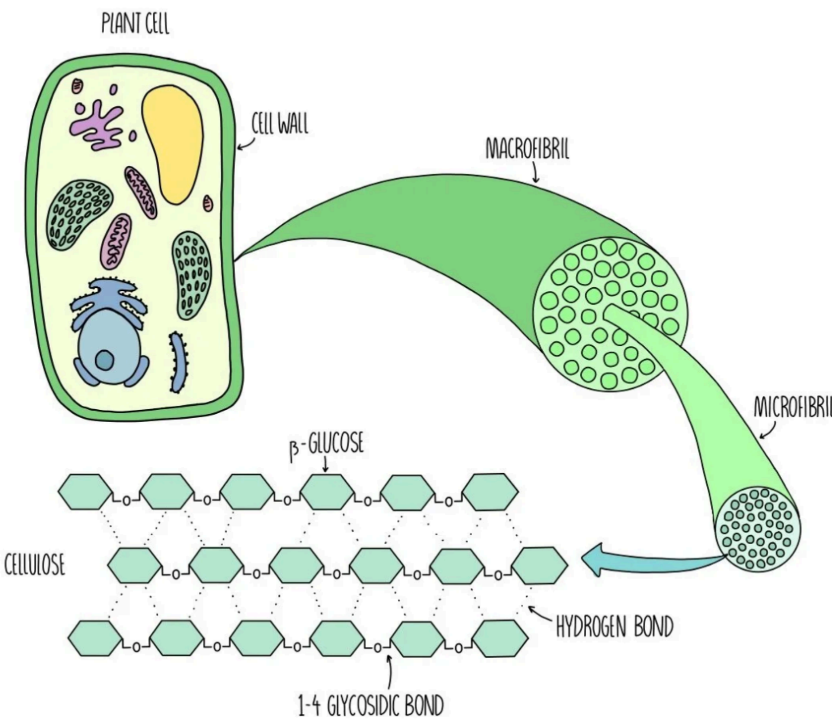
Cellulose

A. **Description**

→ A cellulose molecule is considered to be a **polysaccharide** of **β-Glucose molecules**

→ The most common place to find cellulose is within the cell walls of plant cells

B. **Structure**

  → Due to β-Glucose molecules' characteristic of its right-most OH branch, it has to **flip itself upside down** to be able to form its 1,4-glycosidic bond → In contrast to Amylose (where none of its glucose molecules are flipped), cellulose forms a **straight, uncurled chain** → Many cellulose molecules packed together form **strong hydrogen bonds** which keeps it close to be able to produce microfibrils that form the fibers of the cell walls	

Mentors’ Sources:

- AS & A Level Biology Blue Book
- Save My Exams
- Khan Academy